

An exact spectral backward-error counterexample for LSMR

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

An integer 4×3 least-squares problem has exact LSMR iterates with increasing optimal matrix-only spectral-norm backward error. A spectral completion gives an upper certificate 1979/2000 for the squared first error, and a positive-definite rational matrix gives a strict lower certificate 99/100 for the squared second error. The stated backward-error approximation increases as well. These certificates use exact rational arithmetic, not a floating-point optimizer or substitution of the Frobenius norm.

1 Definitions and exact iterates

For $r = b - Ax$, define

$$\mu(x) = \min\{\|E\|_2 : (A + E)^T((A + E)x - b) = 0\}.$$

Only A is perturbed; the norm is spectral. The approximation uses $\eta = \|r\|_2 / \|x\|_2$, $K = (A^T, \eta I)^T$, $v = (r^T, 0)^T$, and $\tilde{\mu}(x) = \|KK^\dagger v\|_2 / \|x\|_2$. For nonzero x, r , projection algebra gives

$$\tilde{\mu}(x)^2 = r^T A (\|x\|_2^2 A^T A + \|r\|_2^2 I)^{-1} A^T r.$$

Use no damping, $x_0 = 0$, and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{pmatrix}, \quad b = (11, 1, 1, 1)^T.$$

The matrix has full column rank. Put $H = A^T A = \text{diag}(1, 36, 25)$ and $g = A^T b = (11, 6, 5)^T$. Exact LSMR minimizes $\|g - Hx\|_2$ over $\mathcal{K}_k(H, g)$. Its first two iterates are

$$x_1 = \frac{1021}{31201} g = \frac{1}{31201} (11231, 6126, 5105)^T,$$

$$x_2 = \frac{16321g - 383Hg}{110438} = \frac{1}{55219} (87659, 7599, 16865)^T.$$

For $V_1 = (g)$ and $V_2 = (g, Hg)$, direct multiplication checks $x_k \in \text{range } V_k$ and $(HV_k)^T(g - Hx_k) = 0$. The columns of HV_k are independent, so these conditions uniquely certify the minimizing iterates. The third is $(11, 1/6, 1/5)^T$, the exact least-squares solution.

The first residuals are

$$r_1 = \frac{1}{31201} (331980, -5555, 5676, 31201)^T, \quad r_2 = \frac{1}{55219} (519750, 9625, -29106, 55219)^T.$$

The normal residuals decrease, as required by LSMR:

$$\|A^T r_1\|_2^2 = \frac{3593700}{31201} > \frac{5336100}{55219} = \|A^T r_2\|_2^2.$$

2 General lower and upper certificates

For nonzero x , write $s = \|x\|_2^2$, $z = Ax$, $r = b - z$, and

$$C = AA^T, \quad D = \frac{\|r\|_2^2 I - rr^T + zz^T}{s}.$$

Lemma 1. *If $m > n$ and $0 \leq t \leq 1$, then $\mu(x)^2 \geq \lambda_{\min}(tC + (1-t)D)$.*

Proof. For a feasible E whose new residual is nonzero, let u be its unit direction. Then $(A + E)^T u = 0$, so $\|E\|_2^2 \geq u^T C u$. The residual is $uu^T b$, because it is orthogonal to $(A + E)x$. Therefore $Ex = r - uu^T b$, and $\|E\|_2^2 \geq \|r - uu^T b\|_2^2 / s = u^T D u$. A convex combination proves the claim.

For zero new residual, $Ex = r$, so $\|E\|_2^2 \geq \|r\|_2^2 / s$. Choose unit $u \in \ker A^T$, possible since $m > n$. Then $u^T z = 0$ and

$$u^T [tC + (1-t)D] u = (1-t)(\|r\|_2^2 - (u^T r)^2) / s \leq \|r\|_2^2 / s.$$

This proves the same bound in that case. $\square$

Lemma 2. *If a unit vector u has $u^T C u < \kappa$ and $u^T D u < \kappa$, with $\kappa > 0$, then a feasible perturbation has $\|E\|_2 \leq \sqrt{\kappa}$.*

Proof. Let $e = x/\sqrt{s}$ and prescribe

$$E^T u = -A^T u =: de + a_\perp, \quad Ee = (r - uu^T b)/\sqrt{s} =: du + c_\perp,$$

where $a_\perp \perp e$, $c_\perp \perp u$, and the shared scalar is $d = -u^T Ax/\sqrt{s}$. The two squared lengths are $u^T C u, u^T D u$. A completion is

$$E = due^T + ua_\perp^T + c_\perp e^T - \frac{d}{\kappa - d^2} c_\perp a_\perp^T.$$

To bound its norm, put $d_0 = d/\sqrt{\kappa}$, $r_0 = \sqrt{1 - d_0^2}$. In decompositions along u, e , $E/\sqrt{\kappa}$ factors as

$$\begin{pmatrix} 1 & 0 \\ 0 & c_\perp/(\sqrt{\kappa}r_0) \end{pmatrix} \begin{pmatrix} d_0 & r_0 \\ r_0 & -d_0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & a_\perp^T/(\sqrt{\kappa}r_0) \end{pmatrix}.$$

The outside maps have norm at most one by the prescribed length bounds, and the middle map is orthogonal. The prescriptions give $(A + E)^T u = 0$ and $(A + E)x = (I - uu^T)b$, proving feasibility. $\square$

3 Strict increase

For x_1 , choose $w = (250, -1, 1, 27)^T$, $u = w/\sqrt{63231}$, $\kappa = 1979/2000$. Exact evaluation gives

$$u^T C u = \frac{62561}{63231} < \kappa, \quad u^T D u = \frac{1642993919237}{1713779258646} < \kappa.$$

The upper lemma applies. For x_2 , set $t = 5/6$ and $c = 99/100$. Then $tC + (1-t)D - cI = K/2407881992100$, where

$$K = \begin{pmatrix} 206417059721 & -50293465200 & 1125984433750 & -1435003762500 \\ -50293465200 & 83658415217471 & 206242965000 & -26574143750 \\ 1125984433750 & 206242965000 & 61800032332121 & 80360210700 \\ -1435003762500 & -26574143750 & 80360210700 & 11170189945871 \end{pmatrix}.$$

The four leading principal determinants are

$$\begin{aligned} &206417059721, \\ &17265994657467102998545591, \\ &960941324740480331793743845178086291011, \\ &65442104145157248520714038046591467785805073713081. \end{aligned}$$

All are positive. Sylvester's criterion and the lower lemma give

$$\mu(x_1)^2 \leq \frac{1979}{2000} < \frac{99}{100} < \mu(x_2)^2.$$

4 Approximation and rational witness

The squared approximations are exactly

$$\begin{aligned} \tilde{\mu}(x_1)^2 &= \frac{69694107852573439503892031925}{69323394392991282508138323472}, \\ \tilde{\mu}(x_2)^2 &= \frac{5430772101137459612205263871781350}{5387955615790281743396033884265233}. \end{aligned}$$

Cross multiplication yields

$$\tilde{\mu}(x_1)^2 < 1.006 < 1.007 < \tilde{\mu}(x_2)^2,$$

where the decimals are exact rational cutoffs.

The upper completion also has a fully rational form. Put $\omega = w^T w$, $h = w^T z$, $c_0 = r - w(w^T r)/\omega$, $a_0 = -A^T w + (h/s)x$. Then

$$E = -ww^T A/\omega + c_0 x^T/s + \frac{h c_0 a_0^T}{\omega s \kappa - h^2}.$$

The verifier directly checks the perturbed normal equations and positivity of all leading principal minors of $\kappa I - E^T E$. The complete rational matrix is saved in its results file.

A dense variant multiplies A, b by the Hadamard matrix with rows $(1, 1, 1, 1)$, $(1, -1, 1, -1)$, $(1, 1, -1, -1)$, $(1, -1, -1, 1)$, producing

$$A' = \begin{pmatrix} 1 & 6 & 5 \\ 1 & -6 & 5 \\ 1 & 6 & -5 \\ 1 & -6 & -5 \end{pmatrix}, \quad b' = (14, 10, 10, 10)^T.$$

This multiplier is twice orthogonal. Iterates are unchanged and both error quantities double, preserving the strict increases without a zero row.

5 Sources and scope

Recovered references: OpenProblemsInNLA, provisionally IE-17, *Monotonic optimal backward error along LSMR*; D. C.-L. Fong and M. A. Saunders, *LSMR: An Iterative Algorithm for Sparse Least-Squares Problems*, SIAM J. Sci. Comput. 33(5), 2011, 2950–2971, DOI 10.1137/10079687X.¹ The target addressed here uses the explicit spectral-norm definition in Section 1. The current repository statement was not independently retrieved during reconstruction. A different norm or a perturbation model allowing changes to b is outside this certificate's stated claim.

¹<https://arxiv.org/abs/1006.0758>. Reference metadata was retained from earlier notes; source conventions and priority require verification.